

B.Tech III Year II Semester (R20) Supplementary Examinations November/December 2024

DEEP LEARNING

(Artificial Intelligence and Machine Learning)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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|---|----|
| (a) What are the basic components of linear algebra? | 2M |
| (b) Define Eigen decomposition. | 2M |
| (c) What is under fitting in machine learning? | 2M |
| (d) What is the significance of back-propagation in deep feed forward networks? | 2M |
| (e) What is dataset augmentation? | 2M |
| (f) Define semi-supervised learning. | 2M |
| (g) What is the convolution operation in convolutional networks? | 2M |
| (h) What is the efficient convolution algorithms used in practice? | 2M |
| (i) What are recurrent neural networks (RNNs)? | 2M |
| (j) Define bidirectional RNNs. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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|-----------|--|----|
| 2 | (a) How do matrices and tensors differ from each other? | 5M |
| | (b) Explain how singular value decomposition is used in principal components analysis. | 5M |
| OR | | |
| 3 | (a) Compare and contrast conditional probability and marginal probability. | 5M |
| | (b) What are the different types of norms used in linear algebra? | 5M |
| 4 | (a) Discuss the process of learning XOR in deep feed forward networks. | 5M |
| | (b) Explain how bias and variance impact the performance of machine learning models. | 5M |
| OR | | |
| 5 | (a) Compare and contrast maximum likelihood and Bayesian statistics. | 5M |
| | (b) Discuss the challenges that motivate the adoption of deep learning techniques. | 5M |
| 6 | (a) Compare and contrast bagging and ensemble methods. | 5M |
| | (b) Discuss the challenges faced in neural network optimization. | 5M |
| OR | | |
| 7 | (a) Discuss the different regularization techniques used in deep learning. | 5M |
| | (b) Explain the concept of parameter tying and sharing in regularization. | 5M |
| 8 | (a) Discuss the role of convolutional networks in deep learning. | 5M |
| | (b) Describe random or unsupervised features in convolutional networks. | 5M |
| OR | | |
| 9 | (a) What are the applications of convolutional networks in real-world scenarios? | 5M |
| | (b) Explain the importance of basis for convolutional networks in feature extraction. | 5M |
| 10 | (a) Explain the concept of auto encoders in sequence modeling. | 5M |
| | (b) Describe the optimization techniques used for long-term dependencies in sequence modeling. | 5M |
| OR | | |
| 11 | (a) What are the applications of sequence modeling in natural language processing? | 5M |
| | (b) Explain the concept of recursive neural networks and their applications. | 5M |
